ELEC-G001-A | Answers

Teacher copy | Suggested marking | 8 marks

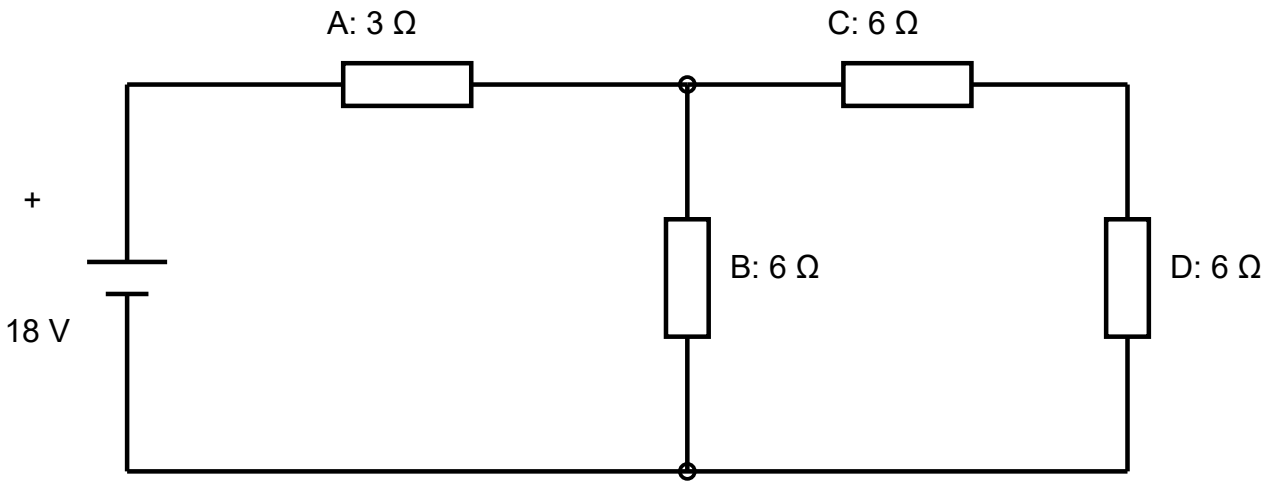


Figure 1

Question 1 — C and D give 12 Ω. B || (C + D) = 4.0 Ω (1 mark). Including A gives 7.0 Ω (1 mark).

Question 2 — Battery current = 18/7 A; parallel-section voltage = 72/7 V (1 mark for a valid method). I_B = (72/7)/6 = 12/7 = 1.71 A (1 mark).

Question 3 — I_A = 18/7 A (1 mark for correct method). V_A = (18/7) × 3 = 54/7 = 7.71 V (1 mark).

Question 4 — I_D = (72/7)/12 = 6/7 A (1 mark). $P_D = (6/7)^2 \times 6 = 216/49 = 4.41$ W (1 mark).

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ELEC-G001-B | Answers

Teacher copy | Suggested marking | 10 marks

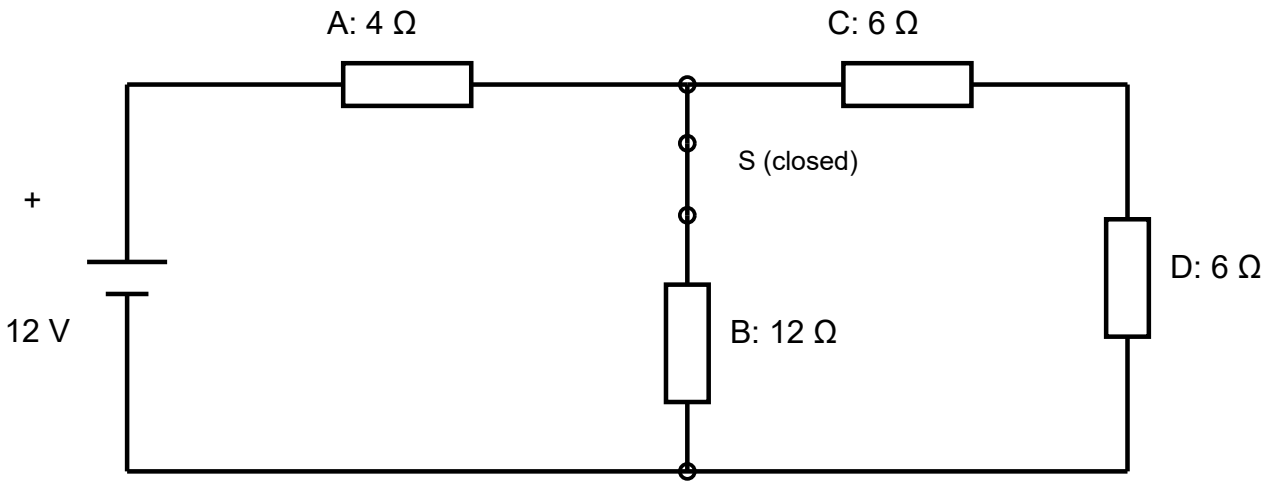


Figure 1

Question 1 — C + D = 12 Ω; B || (C + D) = 6.0 Ω (1 mark). Total resistance = 4 + 6 = 10 Ω (1 mark).

Question 2 — Battery current = 12/10 = 1.2 A. The two branches each have 12 Ω, so they share the current equally (1 mark). I_B = 0.60 A (1 mark).

Question 3 — A carries 1.2 A (1 mark for method). $V_A = 1.2 \times 4 = 4.8$ V (1 mark).

Question 4 — Initially I_D = 0.60 A, so $P_D = 0.60^2 \times 6 = 2.16$ W (1 mark). With B disconnected, A, C and D form a series circuit: I = 12/16 = 0.75 A (1 mark). New $P_D = 0.75^2 \times 6 = 3.375$ W, or 3.38 W (1 mark). The current no longer splits at B: current in D increases from 0.60 to 0.75 A although battery current falls from 1.2 to 0.75 A (1 mark).

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ELEC-G002-B | Answers

Teacher copy | Suggested marking | 4 marks

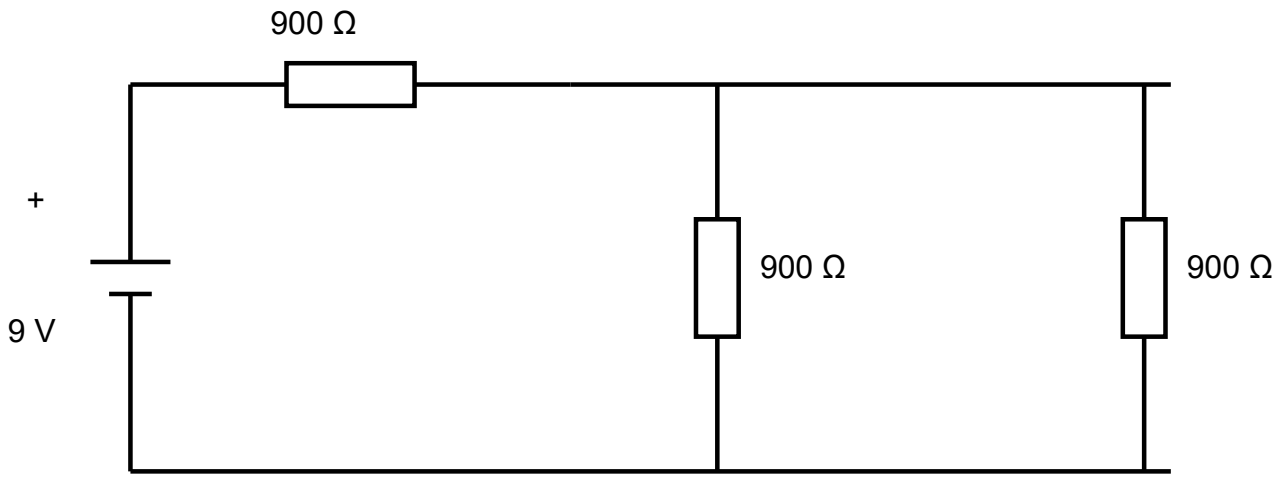


Figure 1

(a) — A is directly across the battery, so V_A = 9.0 V (1 mark). B and C share 9.0 V equally, so each has 4.5 V (1 mark).

(b) — Put two 900 Ω resistors in parallel (450 Ω), in series with the third (1 mark). Total = 1350 Ω, giving 6.0 V across the single resistor and 3.0 V across each parallel resistor (1 mark). See the solution diagram.

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ELEC-G002-A | Answers

Teacher copy | Suggested marking | 4 marks

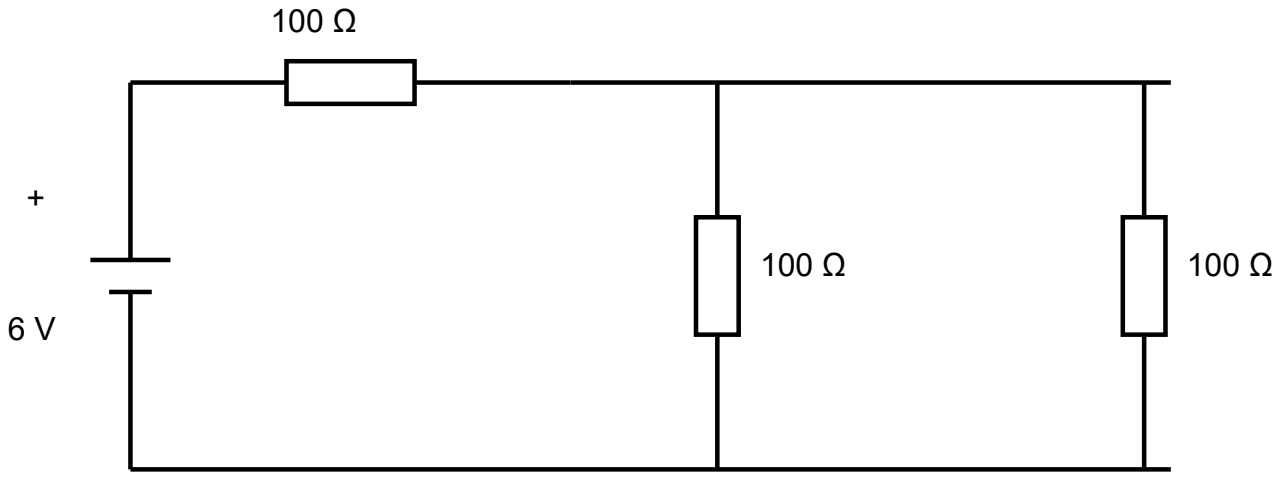


Figure 1

(a) — Connect two 100 Ω resistors in parallel and place the third 100 Ω resistor in series with this pair (1 mark). Label 4.0 V across the single resistor and 2.0 V across each resistor in the parallel pair (1 mark). See the solution diagram.

(b) — The parallel pair is 50 Ω; total = 150 Ω, giving I = 6.0/150 = 0.040 A = 40 mA (1 mark). The single resistor drops 0.040 × 100 = 4.0 V; the parallel pair drops 0.040 × 50 = 2.0 V, so each resistor in the pair has 2.0 V across it (1 mark).

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